

Jairaj Narendran, PhD

Mechanical Engineer | Experimental Mechanics | Design & Prototyping
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EDUCATION

Boston University, College of Engineering **Boston, MA**
Ph.D. in Materials Science and Engineering May 2026

- Thesis: *Functional Nonlinearities Enable Compliance-Switching in Elastogranular Structures*
- Mentored 15+ students in research skills, instructed 80+ students in electronic test equipment, led a graduate mentorship program pairing 20+ students, and hosted networking events with 35+ students and 10+ companies
- Developed curriculum for and mentored 10+ high school students through the Upward Bound program to give STEM exposure to low-income and first-generation college-bound students

M.S. in Materials Science and Engineering, GPA: 3.34/4.00 January 2026

University of Illinois Urbana-Champaign, Grainger College of Engineering **Champaign, IL**

B.S., Bioengineering, Minor in Materials Science and Engineering, GPA: 3.67/4.00 May 2021

WORK EXPERIENCE

Boston University **Boston, MA**
Graduate Research Fellow September 2021 – May 2026

Compliance-Shifting Elastogranular Shells

- Fabricated silicone-molded elastogranular bilayer shells exhibiting tunable, reversible stiffness (up to 3.5X of an unactuated shell) and increased buckling load (2X) with applications in soft robotics and compliant systems
- Built data processing pipelines in MATLAB to extract performance metrics from dynamic testing data and connect geometric parameters to shell characteristics for more efficient design of load-bearing structures
- Determined mechanics of increased shell stiffness and variable buckling behavior through Python simulations of granular jamming on hemispherical surfaces

Bio-inspired Restoration of Aged Concrete Edifices (BRACE) – in collaboration with Battelle, MIT, SDSM&T, and DARPA

- Worked with research engineers, airfield damage repair (ADR) SMEs, and senior microbiologists to design and validate fiber-confined construction materials for rapid ADR from material selection through large-scale testing
- Led project management for multi-institution effort with 15+ research and government stakeholders across 5 institutions, including monthly and biweekly reporting on key performance indicators, risks, and timeline
- Provided engineering oversight during structure construction and assembly, serving as engineering point of contact for 10+ critical production issues and ensuring processes are executed to specification
- Evaluated material strength per ASTM C39 standards and improved structural performance through iterative experimentation and root-cause analysis, surpassing the 400 PSI performance milestone by over 200%

Grasping Delicate Objects with Kirigami Grippers – BU Ignition Award Startup Fund

- Led the full-cycle development of a robotic kirigami grasping tool, addressing customer needs in target markets (agriculture, medicine) and delivering an 8X improvement in fatigue life and a 6X increase in load capacity
- Developed test protocols and SOPs using design-of-experiments methodologies to quantify mechanical strength, fatigue behavior, and failure modes for the selection of optimal materials and geometries
- Designed mechanical components using CAD (OnShape, SolidWorks) and 3D printing, maintained engineering documentation, and used validation and verification to evaluate product performance

SKILLS

- **Mechanical Design:** CAD (OnShape, SolidWorks), GD&T (ASME Y14.5M), design-for-manufacturing, 3D printing (FDM, SLS), additive manufacturing, machining, laser cutter, silicone molding
- **Engineering Analysis & Problem Solving:** root cause analysis, failure analysis, statistical analysis, design-of-experiments, digital image correlation, dynamic mechanical testing, electronic test equipment
- **Data Analysis & Scripting:** MATLAB, Python, ImageJ, CellProfiler, LabVIEW, Arduino IDE
- **Technical & Office:** Microsoft Office 365 (Excel, Word, PowerPoint, Outlook), Adobe Illustrator, Blender

SELECTED PUBLICATIONS

- **J. Narendran**, A.A. M. Campbell, K. Wood, J. Roman, D.P. Holmes (2026). Knit Sleeves for Load Bearing Elastogranular Structures. *In Preparation*
- **J. Narendran**, P. Limcaoco, S. Bastiy, D.P. Holmes (2026). Compliance-Shifting Elastogranular Shells. *In Preparation*